Hongyi Jing

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Education

University of Southern California — Los Angeles, CA, US
Ph.D. in Computer Science — *Aug 2026 – Present*
Advisor: Prof. Yue Wang

University of Southern California — Los Angeles, CA, US
M.S. in Computer Science, GPA: 3.87/4.0 — *Aug 2024 – May 2026*

Huazhong University of Science and Technology — Wuhan, China
B.S. in Artificial Intelligence, GPA: 3.9/4.0 — *Sep 2020 – Jun 2024*

Research Interests

Robot Learning, Foundation Model, Humanoid Whole-body Loco-manipulation

Research Projects

- **$\Psi_{0.1}$: Steerable Long-Horizon Humanoid VLA Model** — *Apr 2026 – Oct 2026*
  - Developed a hybrid predictor–planner–controller for up to five minutes of autonomous loco-manipulation; across seven tasks, improved novel-scene success by **25%** and cross-task progress by **43%** over the strongest baseline.
  - Designed a **time-conditioned image world model** that predicts future visual goals for unseen instructions and tasks, using an explicit time horizon to stabilize goal-guided execution.
  - Pretrained the planner on **800 h human video** to learn variable-length action planning from current–future frame pairs, then adapted it to whole-body control with **100 h humanoid data**.
- **Ψ_0: Open Foundation Model for Humanoid Loco-Manipulation** — *Oct 2025 – Mar 2026*
  - Developed a **staged training** framework that transfers human-video motion priors to humanoid control, combining VLM pretraining with a flow-based MM-DiT action expert for joint-space prediction.
  - Built a single-operator whole-body teleoperation pipeline and integrated training-time **real-time action chunking** to improve locomotion stability and reduce execution jitter.
  - Achieved **over 40%** higher success than the strongest baseline across eight tasks with **less than 1/10** the pretraining data (800 h human video, 30 h robot data).
- **Humanoid Everyday: Multimodal Humanoid Manipulation Dataset** — *Dec 2024 – Sep 2025*
  - Built a dataset of **10.3k trajectories across 260 tasks**, synchronizing RGB-D, LiDAR, tactile, and language modalities.
  - Benchmarked **seven policies**, spanning imitation learning and VLAs, across diverse manipulation tasks to characterize policy strengths and failure modes in open-world humanoid control.
  - Developed a **cloud-based evaluation platform** for remote policy deployment and standardized real-humanoid testing.

Selected Publications

* indicates equal contribution.

[1] *$\Psi_{0.1}$: A Steerable Humanoid VLA Model for Long-Horizon Loco-Manipulation.*
Hongyi Jing*, Jionghao Wang*, Songlin Wei*, Weiduo Yuan, Liang Heng, Haoyi Lin, Yunwan Fu, Junsoo Kim, Andrew Park, Xiawei Liu, Muchen Xu, Xin Li, Wenping Wang, Di Huang, Yue Wang. *[Under Review]*

[2] *Ψ_0: An Open Foundation Model Towards Universal Humanoid Loco-Manipulation.*
Songlin Wei*, **Hongyi Jing***, Boqian Li*, Zhenyu Zhao*, Jiageng Mao, Zhenhao Ni, Sicheng He, Jie Liu, Xiawei Liu, Kaidi Kang, Sheng Zang, Weiduo Yuan, Marco Pavone, Di Huang, Yue Wang. *[RSS 2026]*
Best Paper Award, 3D-LLM/VLA Workshop, CVPR 2026.

[3] *Humanoid Everyday: A Comprehensive Robotic Dataset for Open-World Humanoid Manipulation.*
Zhenyu Zhao*, **Hongyi Jing***, Xiawei Liu, Jiageng Mao, Abha Jha, Hanwen Yang, Rong Xue, Sergey Zakharov, Vitor Guizilini, Yue Wang. *[ICRA 2026]*
Best Paper Award, Embodied AI Workshop, CVPR 2026.
Oral Presentation, From Data to Decisions Workshop, ICRA 2026.

[4] *WE-Dream: Generalizable Humanoid World-Action Models with Latent-Action Pretraining and Routed Attention.*
Liang Heng*, Boqian Li*, Weiduo Yuan*, **Hongyi Jing**, Songlin Wei, Jionghao Wang, Xiawei Liu, Muchen Xu, Haoyi Lin, Henghui Bao, Haonan Chang, Di Huang, Yue Wang. *[Under Review]*

[5] *Fast SAM 3D Body: Accelerating SAM 3D Body for Real-Time Full-Body Human Mesh Recovery.*
Timing Yang, Sicheng He, **Hongyi Jing**, Jiawei Yang, Zhijian Liu, Chuhang Zou, Yue Wang. *[ECCV 2026]*

[6] *SIMPLE: Simulation-Based Policy Learning and Evaluation for Humanoid Loco-manipulation.*
Songlin Wei*, Zhenhao Ni*, Jie Liu*, Zhenyu Zhao*, Junjie Ye, **Hongyi Jing**, Junkai Xia, Xiawei Liu, Michael Leong, Liang Heng, Di Huang, Yue Wang. *[CoRL 2026]*

Skills

Agent tools: Claude Code, Codex
Programming: Python, C/C++, PyTorch, LeRobot, Matlab
Language: Chinese (Native), English (TOEFL: 112, 29R, 30L, 25S, 28W)